DIVIDED POWER STRUCTURES ON P-ADIC COMPLETIONS

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Descent criterion. lemma-kernel (same chapter, named). Criterion (2)(c) certifies that gamma extends to A/p^e A once p^e A cap I is preserved.
- Limits of divided power rings. lemma-limits (same chapter, named). Identifies the limit (A^^, I^^, gamma^^) in the category of divided power rings.

Source reference. Original reference 0315: focused-seeds/geometry/S9-dpa-extend-to-completion/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07H2: sources/stacks-dpa/pinned-original-source.tex, source lines 825–825. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07GV: sources/stacks-dpa/pinned-original-source.tex, source lines 579–579. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let (A, I, γ) be a divided power ring. Let p be a prime. If p is nilpotent in A/I, then*

(1) *the p-adic completion $A^\wedge = \lim_e A/p^eA$ surjects onto A/I,*
(2) *the kernel of this map is the p-adic completion $I^\wedge$ of I, and*
(3) *each γ_n is continuous for the p-adic topology and extends to $\gamma_n^\wedge : I^\wedge \to I^\wedge$ defining a divided power structure on $I^\wedge$.*

If moreover A is a $\mathbf{Z}_{(p)}$-algebra, then

(4) *for e large enough the ideal $p^eA \subset I$ is preserved by the divided power structure γ and*

$$(A^\wedge, I^\wedge, \gamma^\wedge) = \lim_e (A/p^eA, I/p^eA, \bar{\gamma})$$

in the category of divided power rings.

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Proof. Let $t \geq 1$ be an integer such that $p^t A/I = 0$, i.e., $p^t A \subset I$. The map $A^\wedge \to A/I$ is the composition $A^\wedge \to A/p^t A \to A/I$ which is surjective (for example by Algebra, Lemma 0315). As $p^e I \subset p^e A \cap I \subset p^{e-t} I$ for $e \geq t$ we see that the kernel of the composition $A^\wedge \to A/I$ is the p-adic completion of I. The map γ_n is continuous because

$$\gamma_n(x + p^e y) = \sum\nolimits_{i+j=n} p^{je} \gamma_i(x)\gamma_j(y) = \gamma_n(x) \bmod p^e I$$

by the axioms of a divided power structure. It is clear that the axioms for divided power structures are inherited by the maps $\gamma_n^\wedge$ from the maps γ_n. Finally, to see the last statement say $e > t$. Then $p^e A \subset I$ and $\gamma_1(p^e A) \subset p^e A$ and for $n > 1$ we have

$$\gamma_n(p^e a) = p^n \gamma_n(p^{e-1} a) = \frac{p^n}{n!} p^{n(e-1)} a^n \in p^e A$$

as $p^n/n! \in \mathbf{Z}_{(p)}$ and as $n \geq 2$ and $e \geq 2$ so $n(e-1) \geq e$. This proves that γ extends to $A/p^e A$, see Lemma 07H2. The statement on limits is clear from the construction of limits in the proof of Lemma 07GV. $\square$